

LoRa multifunction vehicle node V2

SHEET INDEX / Open a subsystem for component groups, net labels and review notes

Power names are global; matching signal labels below join hierarchical sheet pins.

Power

BEC_IN
ISENSE
VBAT_SENSE

BEC_IN
ISENSE
VBAT_SENSE

File: Power.kicad_sch

MCU

GNSS_RX_PC11
GNSS_TX_PC10
I2C1_SCL
I2C1_SDA
IMU_INT
ISENSE
LORA_NRST_PA3
LORA_RX_PA1
LORA_TX_PA0
NRST
PPS_PB0
PWM_ESC_PC6
PWM_SERVO_PA8
SWCLK
SWDIO
SWO
USB_DM
USB_DP
VBAT_SENSE
VBUS_SENSE

GNSS_RX_PC11
GNSS_TX_PC10
I2C1_SCL
I2C1_SDA
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PWM_SERVO_PA8
SWCLK
SWDIO
SWO
USB_DM
USB_DP
VBAT_SENSE
VBUS_SENSE

File: MCU.kicad_sch

Radio

LORA_NRST_PA3
LORA_RX_PA1
LORA_TX_PA0

LORA_NRST_PA3
LORA_RX_PA1
LORA_TX_PA0

File: Radio.kicad_sch

Navigation

GNSS_RX_PC11
GNSS_TX_PC10
I2C1_SCL
I2C1_SDA
IMU_INT
PPS_PB0

GNSS_RX_PC11
GNSS_TX_PC10
I2C1_SCL
I2C1_SDA
IMU_INT
PPS_PB0

File: Navigation.kicad_sch

Actuators

BEC_IN
PWM_ESC_PC6
PWM_SERVO_PA8

BEC_IN
PWM_ESC_PC6
PWM_SERVO_PA8

File: Actuators.kicad_sch

Debug

NRST
SWCLK
SWDIO
SWO
USB_DM
USB_DP
VBUS_SENSE

NRST
SWCLK
SWDIO
SWO
USB_DM
USB_DP
VBUS_SENSE

File: Debug.kicad_sch

BUILD PATH: review -> order selected BOM -> inspect/assemble -> verify power -> program over SWD -> functional acceptance

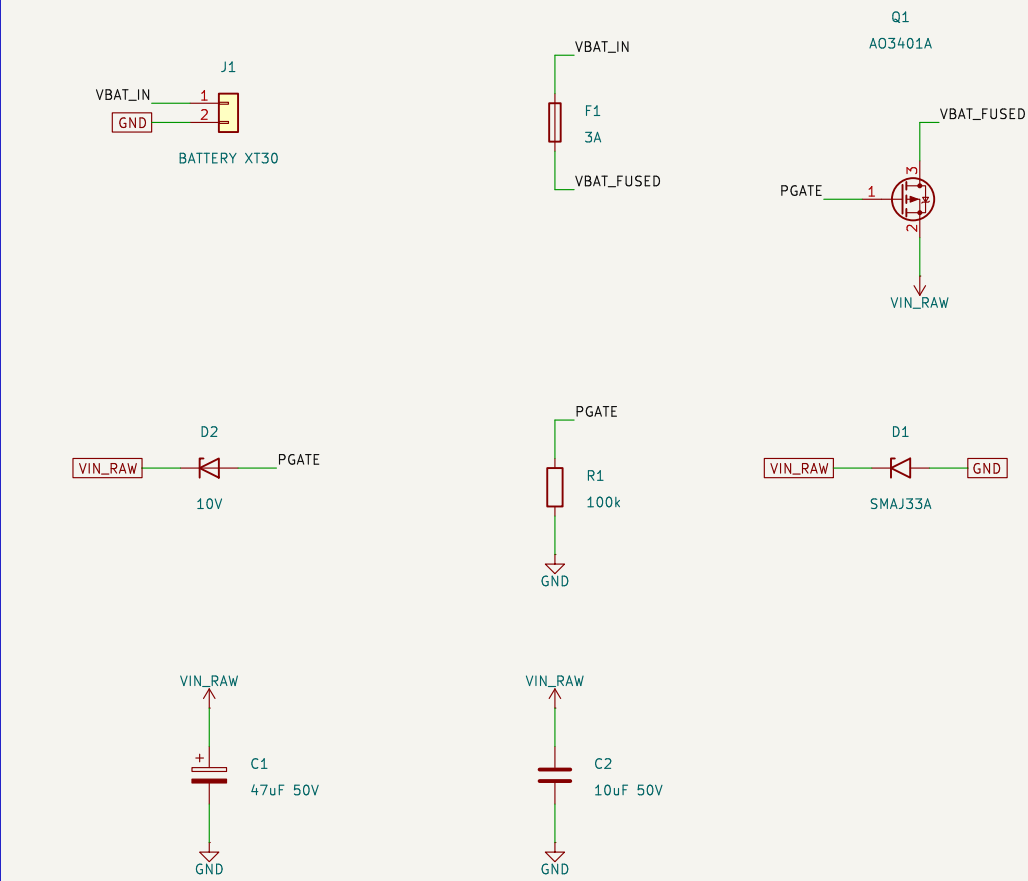
Start: docs/build/README.md | V1 is the existing board; V2 is the multifunction vehicle-node draft.

Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /
File: LoRa_Boat_Controller_V2.kicad_sch

Title: LoRa multifunction vehicle node V2 – Root

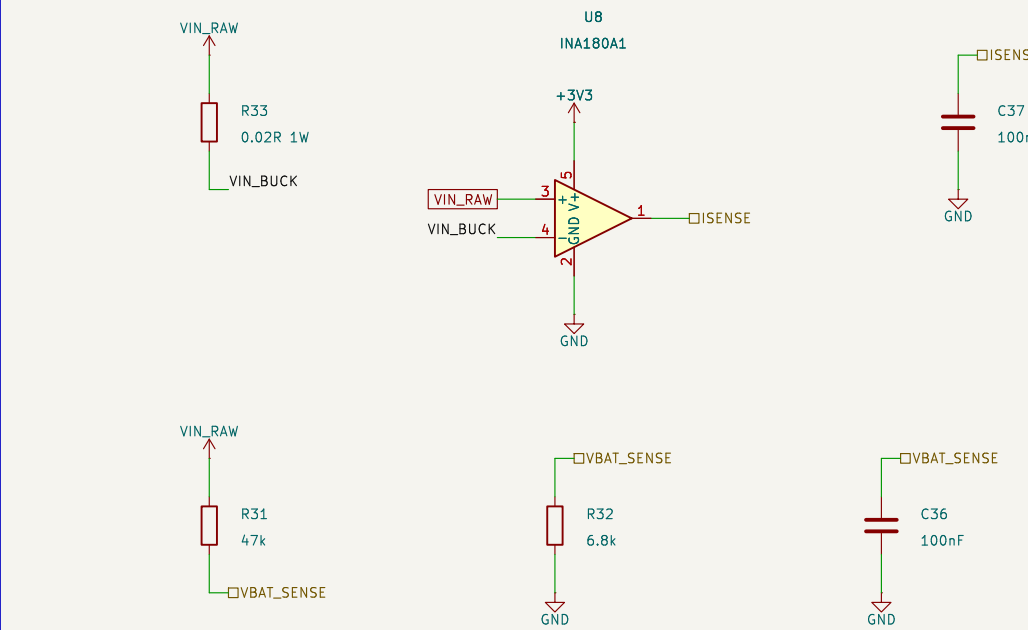
Size: A3 Date: 2026-09-17 Rev: V2 draft
KiCad E.D.A. 9.0.7 Id: 1/7

01 Protected board input



Board supply path only: propulsion battery current bypasses this PCB. Protection ratings and 25–65 margin require design review.

02 Board current and battery sensing



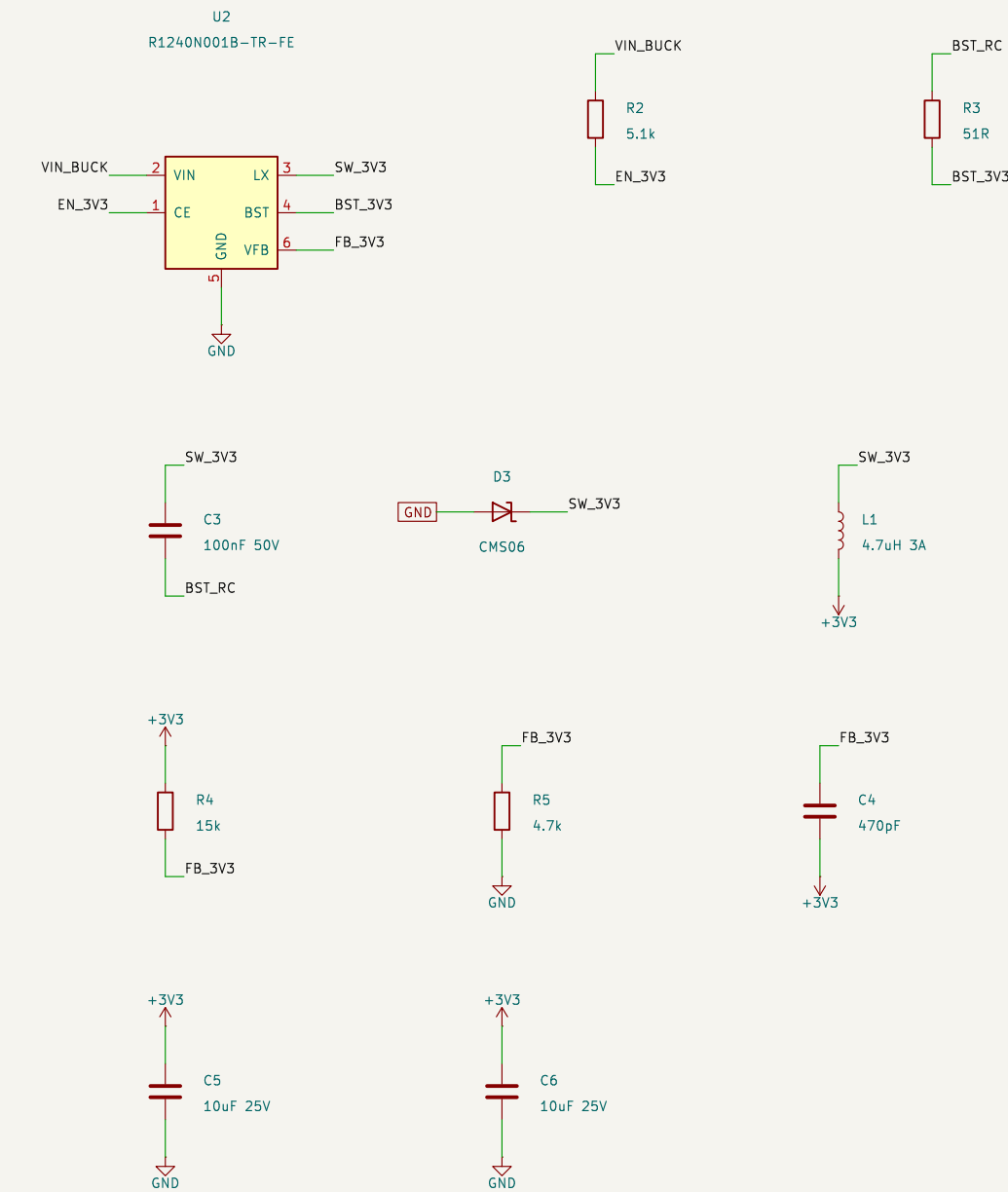
20 mOhm shunt, INA180 gain 20: nominal 0.4 V/A. Divider: 6.8k / (47k + 6.8k). Check ADC scaling before use.



Read named nets within each group: matching labels are electrically connected. Cross-sheet signals use hierarchical ports.

V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

03 3.3 V logic supply



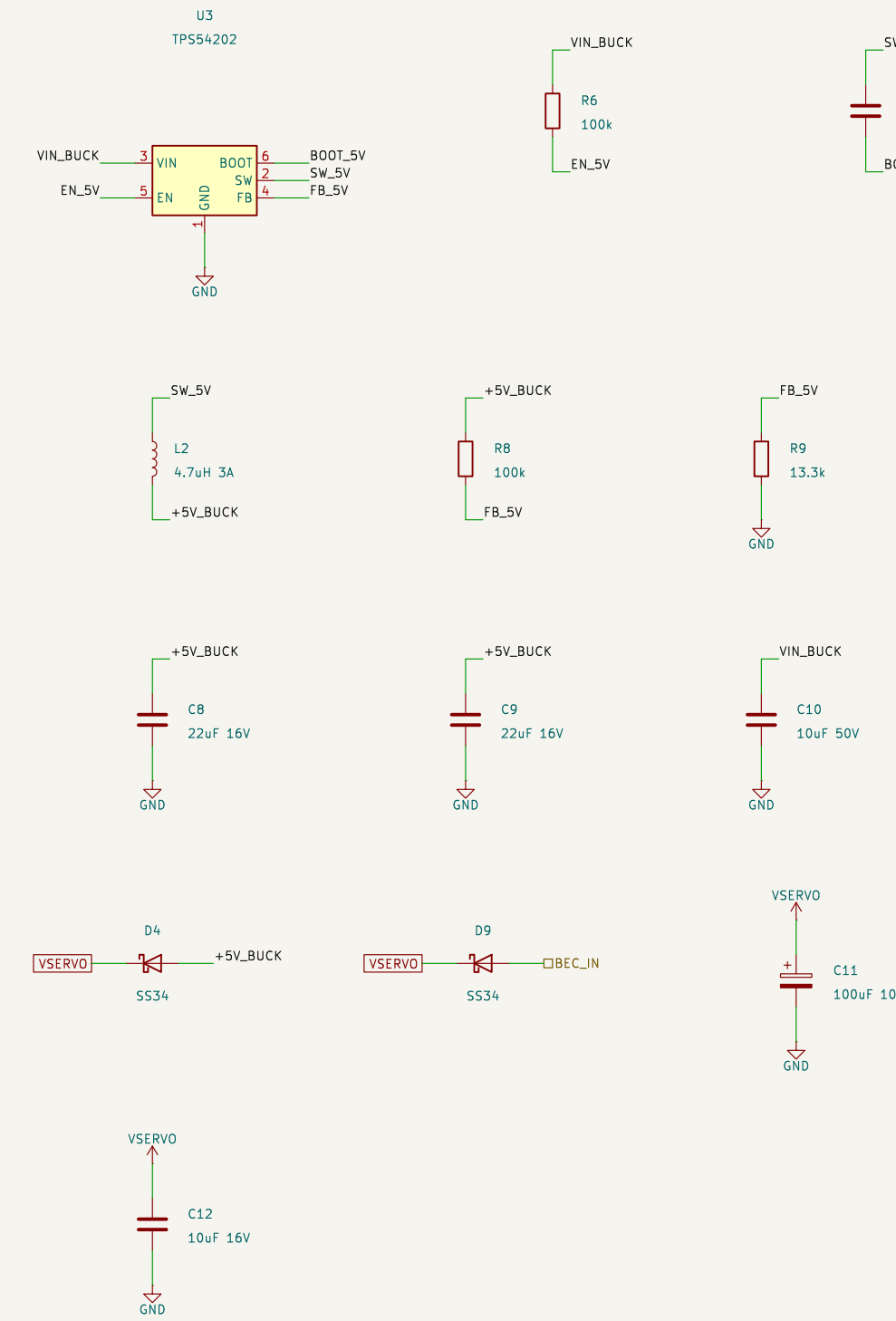
Nominal 3.35 V. Follow the labelled bootstrap, switch and feedback nets; verify component ratings and switching-loop placement before fabrication.

05 Power test points



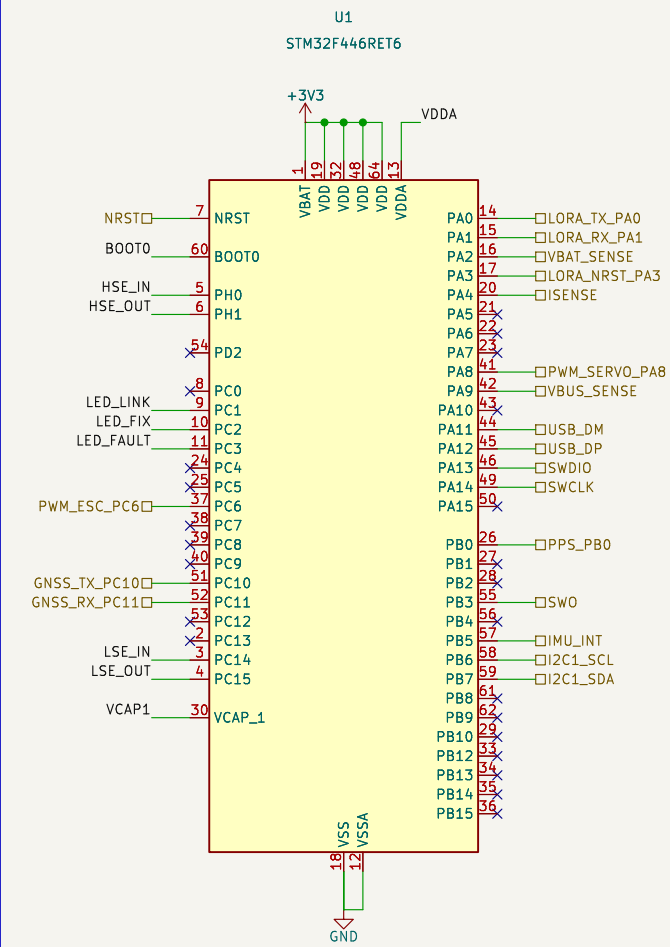
Expected node labels only. No measured rail, ripple, load or thermal qualification is claimed.

04 Actuator supply and BEC isolation



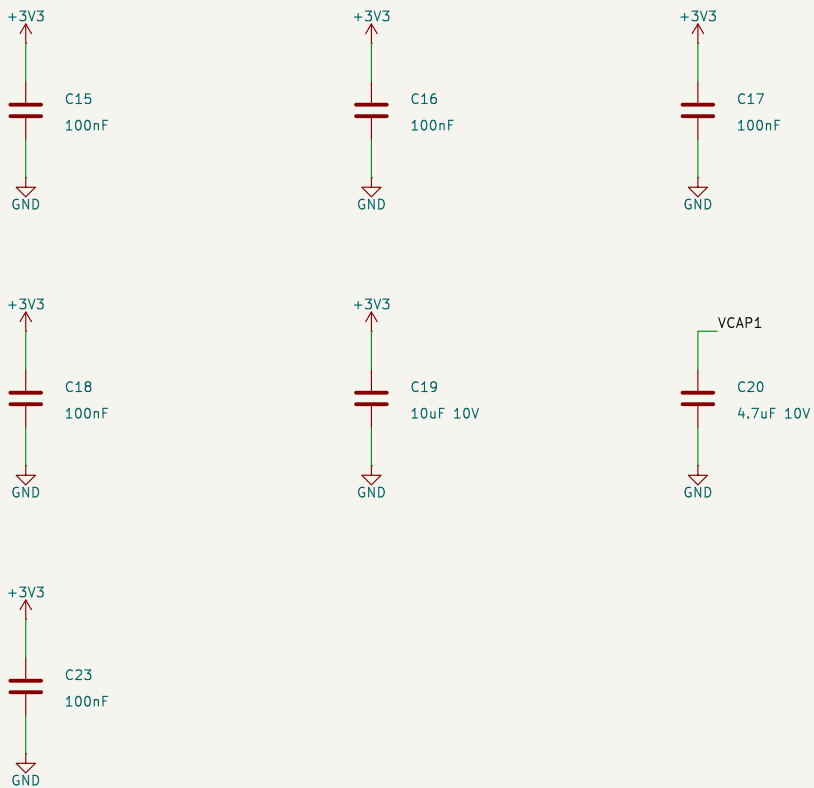
Nominal buck output 5.08 V before OR-ing diode loss. VSERVO is separate from +3V3. Confirm BEC voltage, population option and back-feed behavior.

01 MCU and assigned interfaces



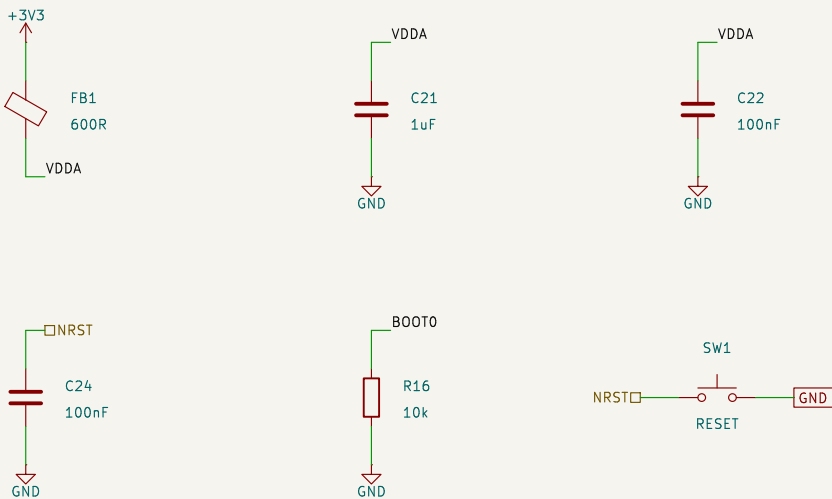
3.3 V logic. Labels define the current board wiring; fleet/ground-vehicle and handheld behavior still requires firmware and interface qualification.

02 Digital and core supply



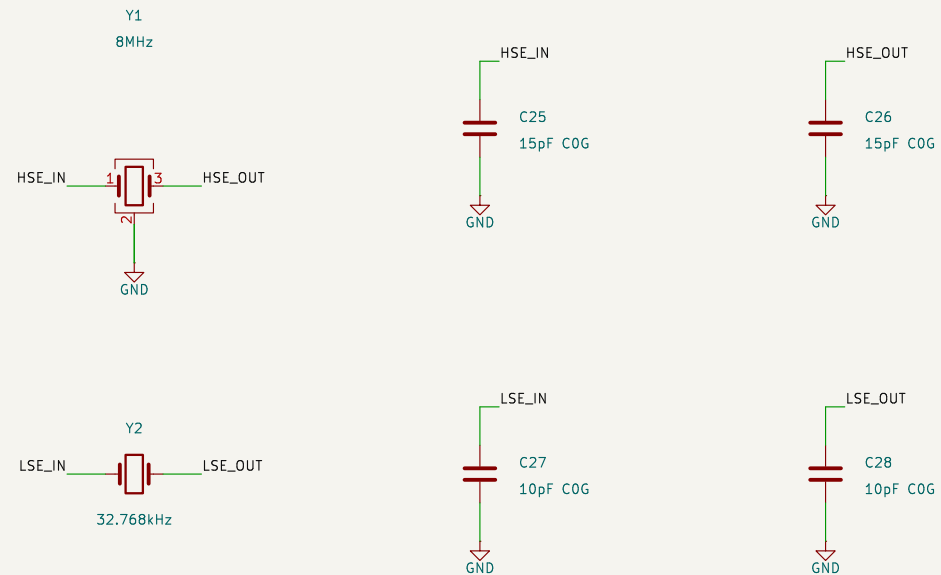
Place each 100 nF capacitor near its VDD pin. C20 is the 4.7 uF VCAP_1-to-GND capacitor. VBAT is tied to VDD.

03 Analog supply and reset defaults



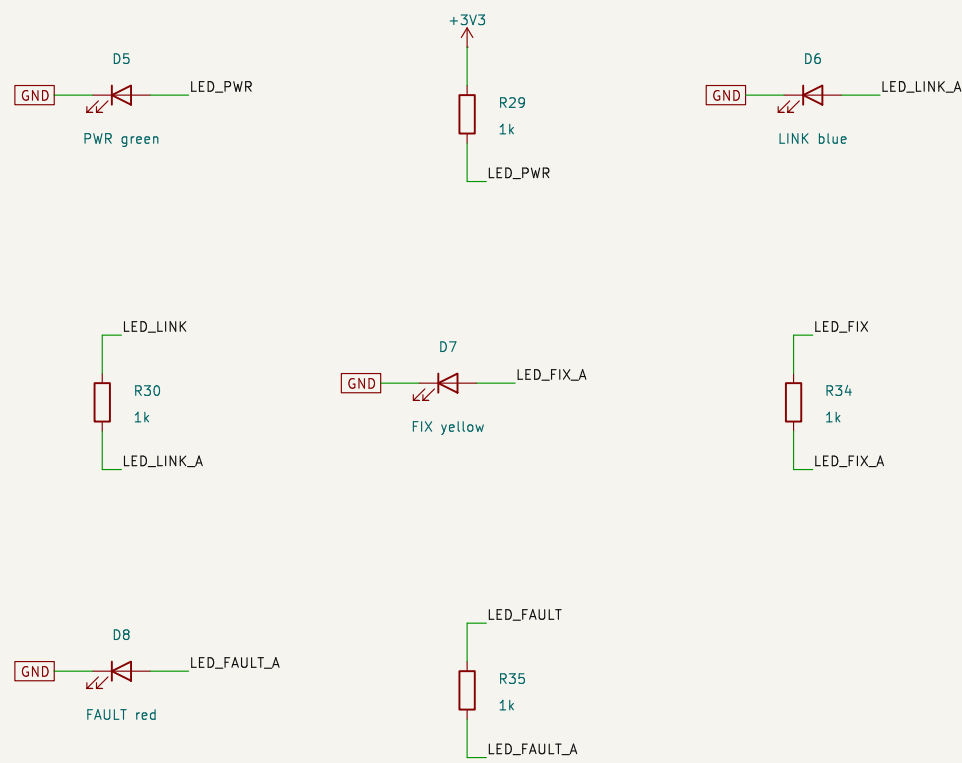
VDDA filter is fed from +3V3. BOOT0 pull-down selects flash boot; NRST is available at the debug header.

04 Clock sources



HSE: 8 MHz / CL 10 pF. LSE: 32.768 kHz / CL 7 pF. Check actual stray capacitance and startup; LSE is a population option.

05 Status indicators



Power, link, fix and fault labels describe intended application states. Firmware must define and verify the indications.



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Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project

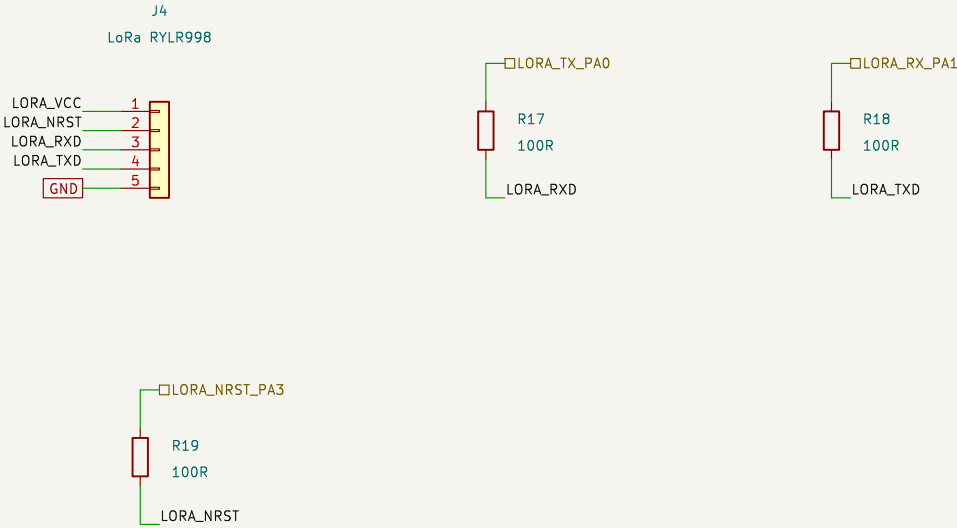
Sheet: /MCU/
File: MCU.kicad_sch

Title: LoRa multifunction vehicle node V2 – MCU

Size: A2 Date: 2026-09-17
KiCad E.D.A. 9.0.7

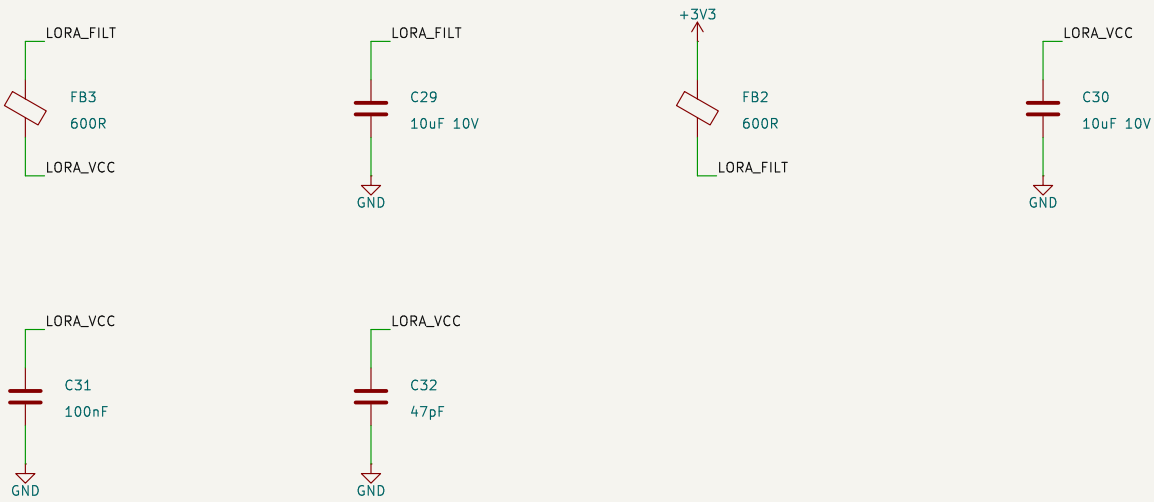
Rev: V2 draft
Id: 3/7

01 Radio socket and serial protection



Verify the chosen REYAX module pinout and voltage. UART/reset series resistors are not voltage translators.

02 Radio supply filtering



Filter returns are to GND. Keep antenna placement clear of switching/actuator noise. Voice/RC airtime coexistence is an open requirement.

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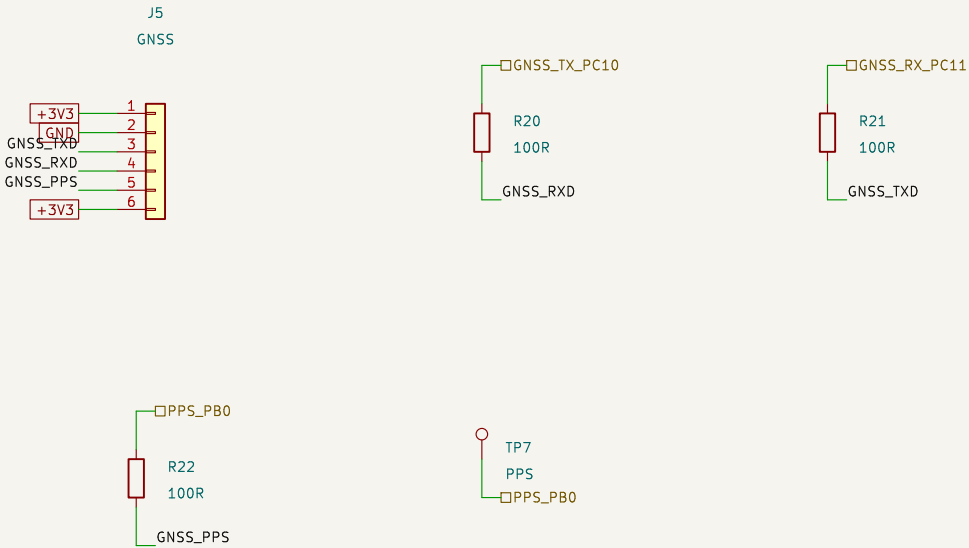
V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /Radio/
File: Radio.kicad_sch

Title: LoRa multifunction vehicle node V2 – Radio

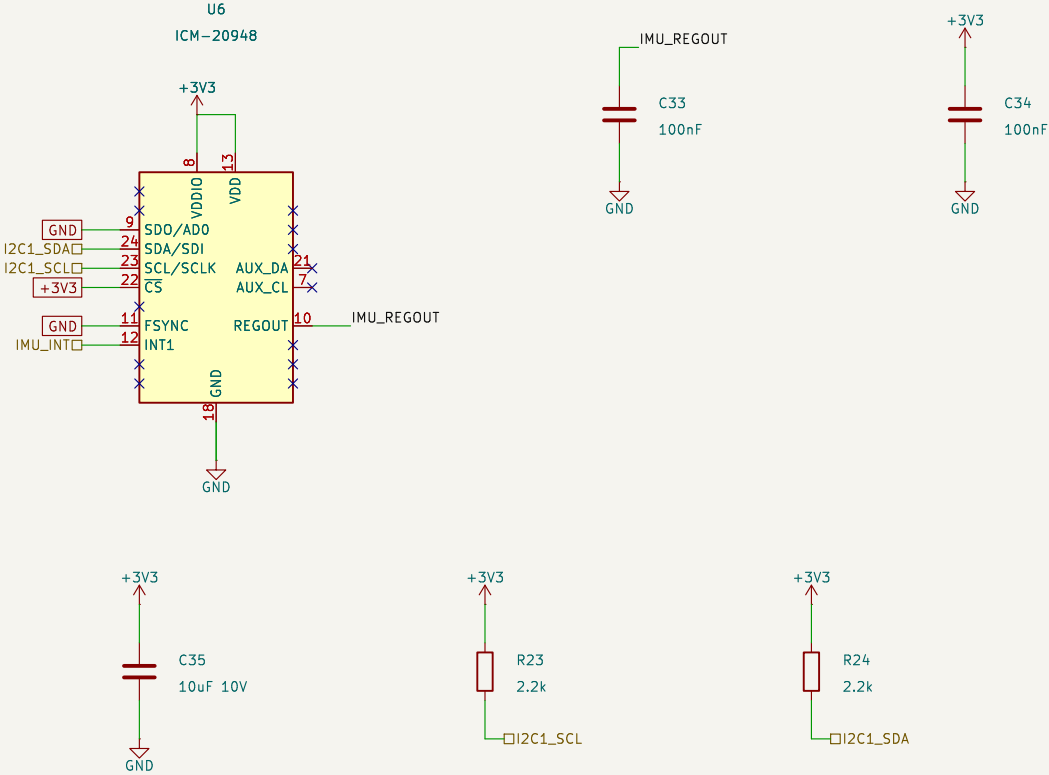
Size: A3	Date: 2026-09-17	Rev: V2 draft
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01 GNSS UART and timing pulse



Check module pin order and PPS polarity before connecting. PBO is intended for PPS capture; it is not the V1 debug-LED output.

02 Inertial sensor



I2C address, VDDIO rating, REGOUT capacitor and AD0/SDO strapping need datasheet review. Draft schematic is not sensor qualification.

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Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /Navigation/
File: Navigation.kicad_sch

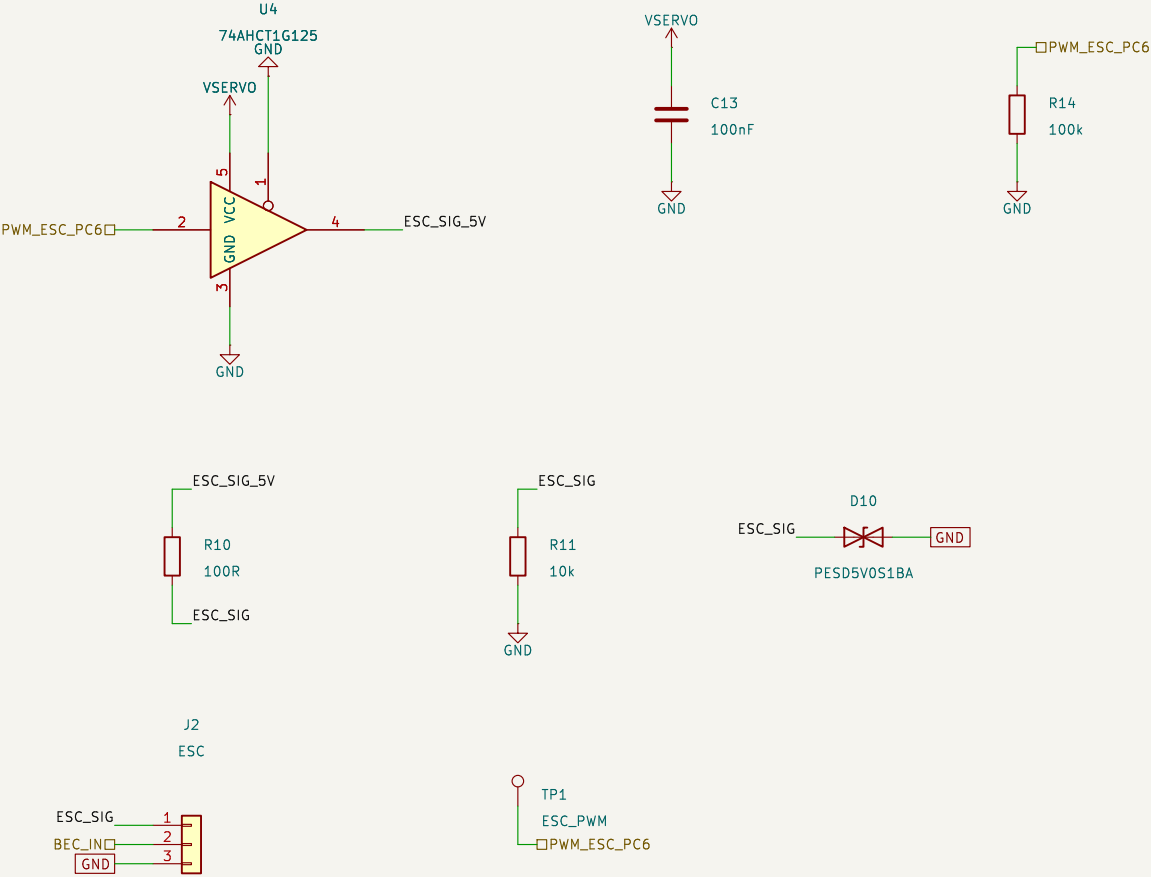
Title: LoRa multifunction vehicle node V2 – Navigation

Size: A3	Date: 2026-09-17	Rev: V2 draft
KiCad E.D.A. 9.0.7		Id: 5/7

V2 / ACTUATORS

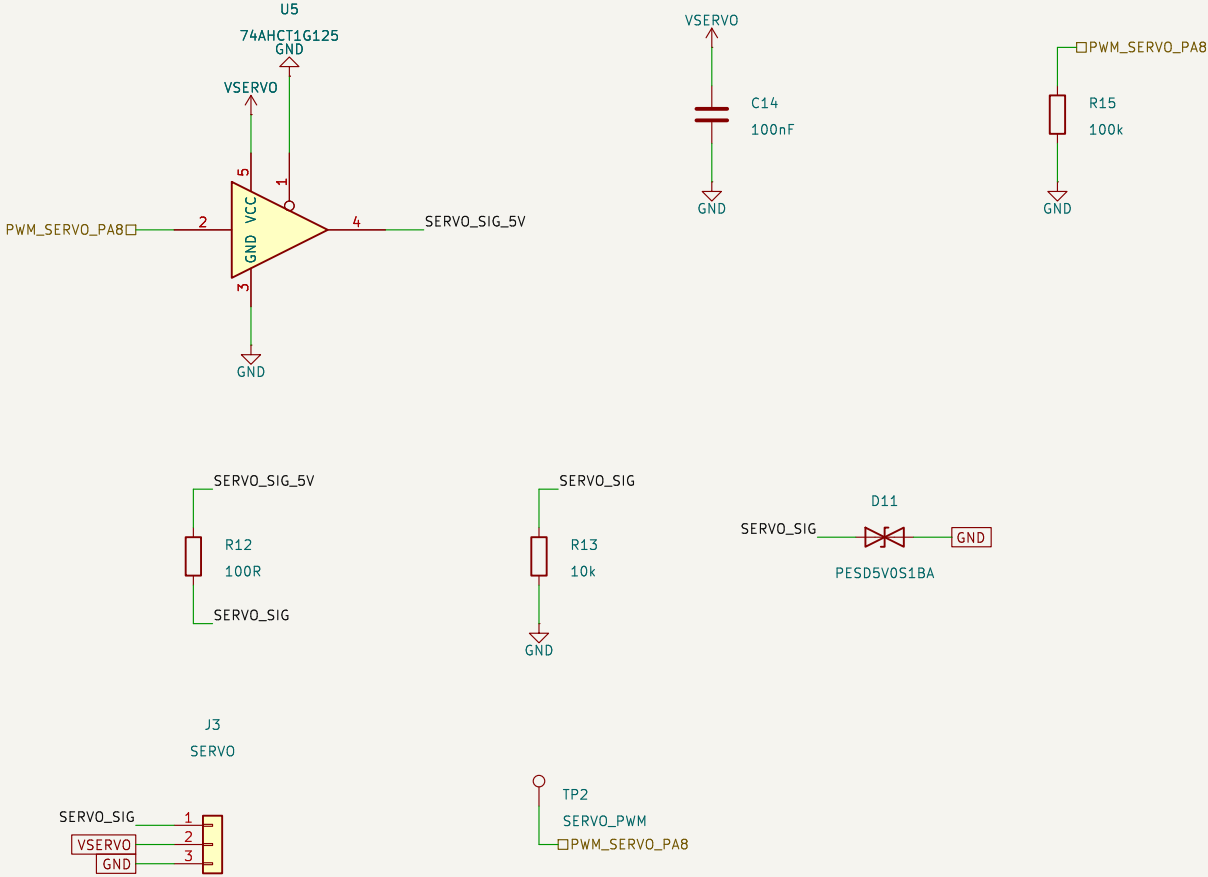
Functional groups / electrical pin numbers / named-net connections

01 Thrust channel



J2: signal / BEC input / GND. MCU-side pull-down does not replace a command-loss failsafe. Verify arm and reset behavior with propulsion disconnected.

02 Steering channel



J3: signal / VESERVO output / GND. Verify buffer supply range against the selected BEC and clamp ratings.

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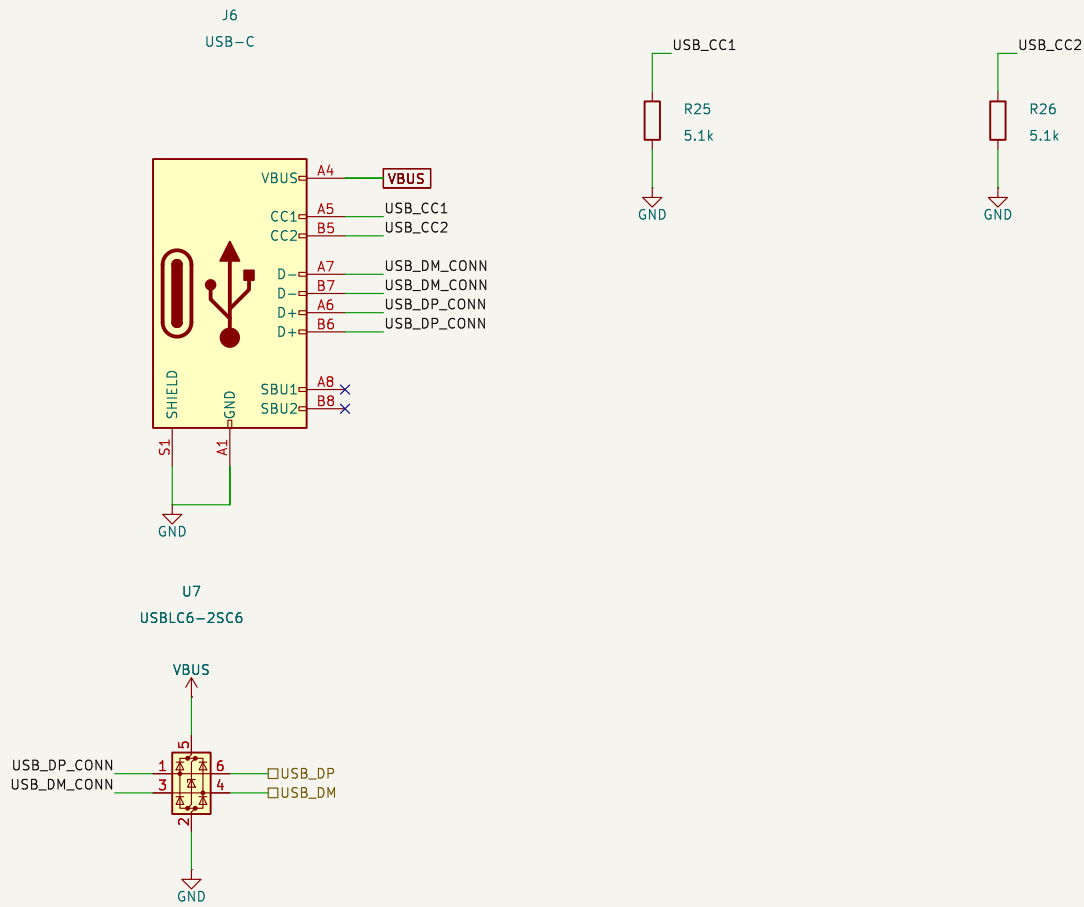
V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /Actuators/
File: Actuators.kicad_sch

Title: LoRa multifunction vehicle node V2 – Actuators

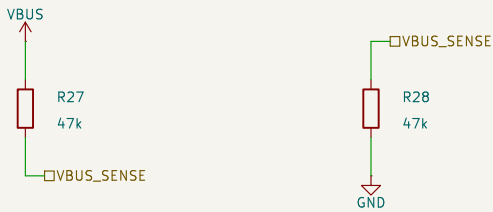
Size: A3	Date: 2026-09-17	Rev: V2 draft
KiCad E.D.A. 9.0.7		Id: 6/7

01 USB-C device connection



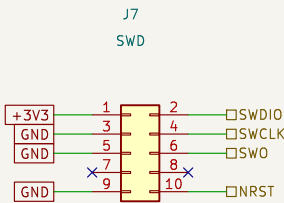
CC pull-downs identify a USB device. Route D+/D- together and place ESD near the connector. USB is not a qualified first-flash path.

02 VBUS detection



47k / 47k divider gives nominal 2.5 V at 5 V VBUS. Firmware must use PA9 consistently with this divided signal.

03 Cortex debug / first flashing



Use SWD for initial programming; check VTref, ground, SWDIO, SWCLK and NRST. See docs/build/FLASHING_V2.md.



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Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /Debug/
File: Debug.kicad_sch

Title: LoRa multifunction vehicle node V2 – Debug

Size: A3	Date: 2026-09-17	Rev: V2 draft
KiCad E.D.A. 9.0.7		Id: 7/7